From Bond-Backed Currency to Real Bills: The Inelasticity Problem and the Origins of the Federal Reserve

Thomas J. Sargent

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Abstract

The National Banking Acts of 1863–1864 created federally chartered banks whose circulating notes had to be backed by holdings of specified U.S. government bonds deposited with the Comptroller of the Currency. Critics from the 1870s onward complained that this arrangement made the currency "inelastic"—unresponsive to the seasonal and cyclical needs of trade. Over the following four decades a range of reform proposals sought to broaden the permissible backing to include railroad bonds, agricultural warehouse receipts, and other non-government assets. The Aldrich–Vreeland Act of 1908, passed in the wake of the Panic of 1907, temporarily allowed emergency notes backed by a wider range of securities and created the National Monetary Commission that would lay the intellectual groundwork for the Federal Reserve. When the Federal Reserve Act was finally passed in December 1913, its long title explicitly promised "to furnish an elastic currency, to afford means of rediscounting commercial paper." The Act fulfilled this promise by embedding the *real bills doctrine* into the Fed's discount window: only short-term, self-liquidating commercial paper—so-called "real bills"—would be eligible for rediscount. We argue that this real bills criterion was no novelty but the revival of an older American tradition—descended from Adam Smith, embodied in the discount practice of the two Banks of the United States and, most purely, in the Louisiana Banking Act of 1842—that the bond-backed national banking system had displaced for half a century. This essay traces the thread from Adam Smith and the antebellum banks through

the Civil War banking statutes, the inelasticity debate, the Aldrich–Vreeland interlude, and finally to the real bills provisions of the 1913 Act, together with the subsequent macroeconomic controversies those provisions engendered—not least the doctrine's fate once the Fed retained Smith's commercial-asset rule but let go of his convertibility anchor.

Contents

1 The National Banking System, 1863–1864

1.1 Origins in the Civil War

Before the Civil War the United States had no uniform national currency and no federal bank-chartering authority. After the demise of the Second Bank of the United States in 1836, banking regulation had devolved almost entirely to the states. The resulting system was fragmented: notes issued by one state's banks circulated at a steep discount elsewhere, and "wildcat banking" scandals in states with lax rules had badly shaken public confidence in paper money.

The exigencies of financing the Union war effort changed the political calculus decisively. Secretary of the Treasury Salmon P. Chase, who had opposed federal involvement in banking before the war, came to see a national currency as indispensable both for fiscal and monetary reasons. Congress responded with the *National Currency Act* of February 25, 1863 (12 Stat. 665, Ch. 58), later renamed the *National Bank Act*, and its more comprehensive successor of June 3, 1864 (13 Stat. 99, Ch. 106).

1.2 The Bond-Backing Requirement

The central mechanism of the new system was straightforward. A bank wishing to issue National Bank Notes had to:

1. Obtain a federal charter from the newly created *Office of the Comptroller of the Currency* (OCC), housed in the Treasury Department.

2. Deposit with the Comptroller a specified quantity of United States government bonds—the 1863 Act required that no less than one-third of the bank's capital take the form of such bonds, held in the Comptroller's vault.

3. Receive in return printed currency (National Bank Notes) up to 90 percent of the par or market value (whichever was lower) of the bonds so deposited.

The notes themselves were uniform in design, printed by the federal government, and identical in appearance across all issuing banks save for a small identification panel. The Comptroller was charged with inspecting banks to verify compliance with the holding requirement—making the OCC the first systematic federal bank supervisor in American history. Hugh McCulloch, the first Comptroller (1863–1865), personally evaluated charter applications, oversaw note design, and arranged for note engraving, printing, and distribution.

State bank notes came under federal attack. A 10 percent federal tax imposed in 1865–1866 on payments made in state bank notes effectively drove them out of circulation, leaving National Bank Notes (and residual Civil War greenbacks) as the dominant hand-to-hand currency. The number of national banks rose from 66 immediately after the 1863 Act to 7,473 by 1913; the number of state banks fell from 1,466 in 1863 to 247 by 1868, before rebounding as bankers exploited the lower capital requirements of state charters and the growing popularity of demand deposits (which were not taxed).

1.3 The Comptroller as Enforcer

That the U.S. Comptroller of the Currency was established in large part to inspect and verify compliance with the bond-deposit requirement deserves emphasis. The Comptroller examined national banks to confirm that the bonds pledged as backing for outstanding notes were actually present and of sufficient value. This supervisory function was novel: previously there was no federal authority empowered to walk into a bank and verify its balance sheet. For our purposes the key point is that the legitimacy of every dollar of National Bank Note currency depended on there being a corresponding U.S. government bond sitting in the Comptroller's vault in Washington.

2 A Rival Principle: Real Bills, Adam Smith, and Antebellum Banking

The bond-backed currency of 1863 embodied one answer to the question of what a bank's notes should be secured by. It was not the only answer Americans had entertained. An older tradition, descended from Adam Smith, held that bank money should be backed not by a pledged *stock* of government securities but by a revolving *flow* of short-term commercial claims—"real bills"—arising from goods actually in production and trade. That tradition governed much of antebellum banking practice, received its purest statutory embodiment in the Louisiana Banking Act of 1842, and, after a half-century of eclipse under the bond-collateral principle, would resurface in the Federal Reserve Act of 1913. To see the 1913 settlement whole, one must first recover the rival principle it revived.

2.1 Adam Smith's Real Bills Doctrine and Its American Reception

Smith's account in *The Wealth of Nations* [Smith, 1776, Book II, Ch. 2] turned on a distinction between two kinds of paper. When a bank discounts "a real bill of exchange drawn by a real creditor upon a real debtor," the loan merely advances money the merchant would otherwise have kept idle, and the bill extinguishes itself once the underlying goods are sold; note issue confined to such paper, Smith argued, "can never exceed the value of the gold and silver which would necessarily circulate in the country if there was no paper money." Against these he set "fictitious" bills—the accommodation paper of "drawing and redrawing" to which "unfortunate traders have sometimes recourse when they are upon the brink of bankruptcy"—for which "there was properly no real creditor but the bank which discounted it." Crucially, Smith's system was anchored by *convertibility*: notes "payable upon demand without any condition, and in fact always readily paid as soon as presented" were "in every respect, equal in value to gold and silver money." The real-bills rule was prudent counsel about a bank's assets; it was specie convertibility that ultimately bounded the note issue—a point whose neglect, as we shall see, would prove the doctrine's undoing.

Smith's book reached America quickly and authoritatively. The first American edition appeared in Philadelphia in 1789; Jefferson pronounced it "the best book extant" in political economy, and Hamilton mined it—while disputing its free-trade conclusions—in the *Report on Manufactures*. But it was received as a guidebook for enlightened statesmen rather than a code to be enacted, invoked selectively and bent to American conditions. Its banking doctrine entered American life less through explicit citation than through the everyday practice of the counting-house.

2.2 Real Bills in Practice: The Banks of the United States

Antebellum commercial banking was real-bills banking in form. Most bank loans were *discounts* rather than simple loans—advances against commercial paper, that is bills of exchange and promissory notes—made for short terms, from a few days to six months. The First Bank of the United States (1791–1811) increasingly specialized in exactly this business: its commercial customers rose from about 40 percent of earning assets in 1793 to more than 85 percent by 1809. Under Nicholas Biddle the Second Bank (1816–1836) dealt heavily in bills of exchange—short-dated (sixty- to ninety-day), self-liquidating orders that represented the value of agricultural exports and commercial imports—and disciplined the note issue of state banks by promptly returning their notes for redemption in specie [Hammond, 1957]. The two pillars of Smith's system, self-liquidating commercial paper and specie convertibility, were thus both present in the practice of the early national banks, whether or not their managers reached for the theoretical genealogy.

2.3 The Theorists Preferred Convertibility

It is a revealing feature of the American debate that its most careful writers rested their systems not on the real-bills rule but on the specie anchor behind it. Albert Gallatin's *Considerations on the Currency and Banking System of the United States* [Gallatin, 1831] reasoned in essentially quantity-theoretic, hard-money terms: paper should never be allowed to exceed the specie it displaced. Condy Raguet's *Treatise on Currency and Banking* [Raguet, 1839] made prompt convertibility into specie the sole test of a sound bank, distinguishing banks of deposit from banks of circulation that "lend their credit." Neither built his system on Smith's real-bills criterion; both located monetary discipline in redemption. The lesson—which bears directly on the question of *whether* the American doctrine was truly Smithian—is that the real-bills principle lived in American bank *practice* more than in American monetary *theory*, where convertibility, not the character of bank assets, was understood to be the binding constraint.

2.4 Real Bills Made Law: The Louisiana Banking Act of 1842

The one antebellum statute that fused the two elements of Smith's system into an enforceable rule was the Louisiana Banking Act of 1842, the "Forstall System" designed by the New Orleans financier Edmond Jean Forstall and signed on 5 February 1842. The Act required each bank to hold specie equal to one-third of its combined note and deposit liabilities—the first American reserve requirement to reckon against deposits as well as notes [Carlson, 2013]—while backing the remaining two-thirds with commercial paper "maturing in less than ninety days and not renewable." It separated a bank's long-dated "dead-weight" loans, made against paid-in capital, from the short, self-liquidating "movement" paper that alone could support its demand liabilities, and it curbed precisely the accommodation lending Smith had warned against [Green, 1969, Neu, 1969]. The result was the purest institutional embodiment of the real-bills principle in antebellum America—and, not coincidentally, a durable one: Louisiana's banks were among the few to maintain specie payments through the Panic of 1857, and the system endured until the Civil War. It worked *because* it supplied what Smith's doctrine required, and what looser "real bills" banking omitted: a hard convertibility anchor sitting beneath the commercial-asset rule.

2.5 The Rival Principle: Bond-Collateral Free Banking

Against this commercial-asset tradition stood a second antebellum principle—and it was the second that the National Banking Acts would nationalize. Under the New York Free Banking Act of 1838 and its imitators, a bank secured its notes not with revolving commercial paper but by depositing *state and federal bonds* with a state officer, who would sell the bonds to reimburse noteholders if the bank failed [Bodenhorn, 2000]. The New York statute was, as historians have long noted, a principal model for the National Bank Act of 1863, which simply substituted federal bonds and a federal Comptroller for state bonds and a state treasurer [Rockoff, 1974]. Antebellum America thus held *two* competing answers to the backing question—commercial-asset (real bills), epitomized by Louisiana 1842, and bond-collateral, epitomized by New York 1838. The Civil War resolved the contest, for half a century, in favor of bonds: the bond-deposit requirement of Section 1 was the free-banking principle raised to national scale, and the real-bills principle went dormant in federal law until 1913.

2.6 Whether It Was Really Smith

How much of this owed to Adam Smith? The honest answer is: the sanction more than the substance. As Lloyd Mints observed in the classic critical history that gave the doctrine its name, Smith was its "first thoroughgoing exponent," and real bills became the main reliance of the agitators for banking reform in the United States down to 1913 [Mints, 1945]. Yet antebellum American real-bills banking is best understood as pragmatic commercial practice that found in Smith a prestigious warrant, not a doctrine mechanically deduced from him—the more so because the Americans who theorized most carefully grounded soundness in convertibility rather than in the character of bank assets. And that emphasis was the correct one. There are, in truth, two real-bills doctrines latent in Smith: a sound prudential norm (a bank does well to hold short, self-liquidating paper) and an unsound monetary one (that lending only against real bills automatically yields the right *quantity* of money). The second is mistaken, because the true anti-over-issue anchor is redeemability, not the asset type; where convertibility disciplined it—as in Louisiana—the real-bills rule was harmless and even salutary, and where it did not, "real bills" banking drifted toward over-issue and suspension [Mints, 1945, Humphrey, 1982]. This distinction, latent in the antebellum experience, would return to haunt the Federal Reserve: the 1913 Act revived the commercial-asset principle (Section 5) but eventually operated it without the specie anchor that had made Smith's original coherent—exactly the vulnerability Thornton had identified and the Depression would expose (Sections 5–6).

3 The Inelasticity Problem, 1865–1907

3.1 Why Bond-Backing Created Rigidity

The very feature that gave National Bank Notes their credibility—the requirement that each dollar of notes be backed by a federal bond—also made the currency supply hostage to fiscal policy rather than to the needs of commerce. The total stock of National Bank Notes that could lawfully circulate was bounded above by (roughly) 90 percent times the total par value of the designated government bonds held by all national banks in aggregate. Three structural features made this constraint binding:

1. **Declining bond supply after the Civil War.** The federal government ran large primary surpluses from the mid-1860s through the 1880s and used the proceeds to retire its war debt. As the supply of outstanding eligible bonds fell, the monetary base of National Bank Notes contracted or was capped even as the economy grew.

2. **Premium prices for eligible bonds.** Because holding eligible bonds entitled a bank to issue profitable currency, demand for those bonds pushed their prices *above* par. A bank that paid a premium for a bond received notes at the bond's par value, not its market price, compressing the profit margin of note issue and reducing the incentive to expand note circulation.

3. **Seasonal and geographic rigidities.** The agrarian economy of the South and West needed sharply higher volumes of currency during planting and harvest seasons. Because National Bank Note issuance was tied to bond portfolios—not to seasonal credit demand—banks could not expand circulation when farmers needed cash and contract it again when crops were sold. The result was recurrent seasonal currency shortages and elevated interest rates in interior agricultural districts, punctuated by periodic financial panics.

Friedman and Schwartz [Friedman and Schwartz, 1963] documented that the currency component of the money supply under the National Banking System was indeed far less responsive to economic conditions than one would expect from a well-designed monetary system. Johnson [Johnson, 1999] summarized the pre-Fed banking system as characterized by "immobile reserves and inelastic currency."

3.2 Political Demands for Currency Reform

The inelasticity of the currency became a major political grievance from the 1870s onward. Broadly, two coalitions formed:

Greenbackers and silver inflationists. The most radical critics demanded that the federal government simply issue more fiat money—greenbacks—or that silver be monetized at a generous ratio. William Jennings Bryan's famous "Cross of Gold" speech at the 1896 Democratic Convention crystallized this position. These advocates cared less about the *backing* of currency than about its *quantity*; they wanted the money supply to be large enough to relieve debt-burdened farmers and spur economic activity.

Asset-currency reformers. A more technically focused group, drawn largely from commercial banking circles in the Northeast but with agrarian sympathizers in the Midwest, argued that the problem was specifically the *bond-backed* structure of note issue. Their proposed remedy was *asset currency*: allowing national banks to issue notes backed not only by government bonds but also by a wider range of productive assets. The menu of proposed alternative backing included:

- *Railroad and corporate bonds.* By the 1880s railroad bonds vastly exceeded federal bonds in total outstanding volume; asset-currency advocates argued that high-grade railroad paper was just as safe and far more plentiful.

- *Agricultural warehouse receipts.* Farmers' groups and some midwestern bankers proposed that banks should be able to issue notes against warehouse receipts for stored grain, cotton, and other commodities. This would make the currency supply directly responsive to the seasonal rhythm of agriculture.

- *General "good assets."* Some proposals envisaged allowing note issue against any sound asset on a bank's books, subject to regulatory approval—a forerunner of the broader "eligible paper" concept that would enter the Federal Reserve Act.

The common thread in asset-currency proposals was the complaint that the existing system was "too inelastic," that it prevented currency from being "responsive to the needs of trade." This phrase—*responsiveness to the needs of trade*— would recur across decades of monetary debate and find its ultimate echo in both the Aldrich–Vreeland Act and the Federal Reserve Act.

3.3 Notable Reform Proposals and Legislation

Successive Secretaries of the Treasury and various congressional committees floated asset-currency bills through the 1880s and 1890s. The Indianapolis Monetary Convention of 1897 brought together business and banking representatives who formally endorsed asset currency backed by commercial paper. The American Bankers Association lobbied actively for a reform along these lines throughout the 1890s and early 1900s. None of these proposals passed Congress, stymied by the cross-cutting politics of money: Eastern bankers feared that broadening the acceptable backing would destabilize the currency, while agrarians found asset-currency reform insufficiently generous.

4 The Panic of 1907 and the Aldrich–Vreeland Act of 1908

4.1 The Panic of 1907

The Panic of 1907 brought the defects of the National Banking System into sharp relief. A stock-market crash and a run on several New York trust companies triggered a cascade of bank suspensions. The crisis was arrested only through the personal intervention of J.P. Morgan, who organized a pool of private funds to shore up the threatened institutions—a striking demonstration that in the absence of a lender of last resort, private coordination by a single wealthy financier was the banking system's only circuit-breaker. The episode produced a broad consensus across political lines that the banking and currency system needed fundamental reform.

4.2 The Aldrich–Vreeland Act (May 30, 1908)

Congress responded with unusual speed. The Aldrich–Vreeland Act (Pub. L. 60-169, 35 Stat. 546), introduced by Representative Edward Vreeland (Republican, New York) and championed in the Senate by Senator Nelson W. Aldrich (Republican, Rhode Island), was signed into law by President Theodore Roosevelt on May 30, 1908.

The Act had two main provisions.

Emergency currency. National banks could form *National Currency Associations* in groups of ten or more, with aggregate capital of at least $5 million. Having formed such an association, member banks could issue *emergency currency* backed not merely by government bonds but by "almost any securities the banks were holding"—including railroad and other corporate bonds. Once approved by association officers and the Comptroller of the Currency, these notes could be placed in circulation as legal tender. The emergency notes bore a tax of 5 percent per annum for the first month outstanding, rising by 1 percentage point each subsequent month. This punitive tax was designed to ensure that the emergency currency would be retired promptly once conditions normalized. As it turned out, no notes were issued under this provision during peacetime conditions.

National Monetary Commission. The Act created the *National Monetary Commission* (NMC), chaired by Senator Aldrich, to study banking and currency systems in the United States and abroad and to recommend comprehensive reforms. The NMC commissioned several dozen studies of banking practices in Europe, particularly in France, Germany, and Britain, and its reports became the intellectual foundation on which the Federal Reserve System was constructed.

4.3 The Aldrich–Vreeland Act in Practice: August–October 1914

The emergency currency provisions proved their worth at the outbreak of World War I in August 1914, when foreign creditors—particularly British banks—demanded immediate gold payments that would otherwise have drained the U.S. monetary base. Secretary of the Treasury William Gibbs McAdoo activated the Aldrich–Vreeland machinery, and by October 23, 1914, some $368,616,990 in emergency notes had been placed in circulation, averting a banking panic. The notes were entirely withdrawn once the Federal Reserve banks opened for business in November 1914.

4.4 From Aldrich–Vreeland to the Federal Reserve

The NMC submitted its final report to Congress on January 9, 1912. The report accompanied draft legislation—the *Aldrich Plan*—proposing a centralized National Reserve Association with fifteen regional branches. The Plan called for a lender of last resort that would discount commercial paper for member banks, making the currency supply elastic and eliminating the bond-backing straitjacket. Opposition arose quickly: rural and progressive Democrats feared that the Association would be controlled by Wall Street's "Money Trust." After extensive hearings by the Pujo Committee (May 1912–January 1913) into alleged financial concentration, President-elect Wilson's administration stepped in with a modified plan that preserved the decentralization demanded by the Bryan wing of the Democratic Party.

5 The Federal Reserve Act of 1913 and the Real Bills Doctrine

5.1 The Long Title as a Statement of Intent

The Federal Reserve Act, signed by President Woodrow Wilson on December 23, 1913, bore the official long title:

> *An Act to provide for the establishment of Federal reserve banks, to furnish an elastic currency, to afford means of rediscounting commercial paper, to establish a more effective supervision of banking in the United States, and for other purposes.*

The phrase "to furnish an elastic currency" was a direct answer to the four-decade complaint about the National Banking System. "To afford means of rediscounting commercial paper" signaled the mechanism by which elasticity would be achieved: member banks could bring eligible commercial paper to their regional Federal Reserve bank, which would discount it (lend against it), providing the bank with reserves and thus allowing it to expand its loans and note issue.

5.2 Eligible Paper and the Real Bills Doctrine

What made paper "eligible" for rediscount at the Fed? Section 13 of the Act specified that the Federal Reserve banks could discount "notes, drafts, and bills of exchange arising out of actual commercial transactions." The key adjective was *commercial*: the paper had to represent a real transaction in goods or services, not a speculative financial transaction. Further criteria required that eligible paper be:

- *Short-term*: maturing within 90 days (or six months for agricultural paper);
- *Self-liquidating*: arising from a specific productive transaction whose completion would generate the funds to repay the loan; and
- *Drawn for a commercial purpose*: explicitly excluding paper "issued or drawn for the purpose of carrying or trading in stocks, bonds, or other investment securities."

This set of criteria precisely codified what monetary theorists called the *real bills doctrine* (also known as the *commercial loan theory of banking*).

5.3 Intellectual Origins of the Real Bills Doctrine

The real bills doctrine has roots in seventeenth- and eighteenth-century banking thought. John Law (1705) sketched the core idea—an "output-governed currency secured to real property and responding to the needs of trade." Adam Smith gave the doctrine its classical formulation in *The Wealth of Nations* [Smith, 1776, Book II, Ch. 2], arguing that a bank that confines its loans to the discount of real bills drawn against actual goods in production will never over-issue its notes: as goods are sold and bills repaid, the notes return to the bank and are extinguished, so the money supply automatically tracks the real volume of transactions.

Henry Thornton (1802) identified the doctrine's central flaw in his *An Enquiry into the Nature and Effects of the Paper Credit of Great Britain*: by tying the money supply to the *nominal* (price times quantity) rather than the *real* volume of trade, the doctrine set up a positive feedback loop from prices to money creation to prices. If the central bank accommodates every demand for credit arising from trade conducted at current prices, and if prices are rising, the money supply will expand along with prices—adding fuel to inflation rather than quenching it.

Despite Thornton's critique, the doctrine retained influential adherents. Thomas Tooke and John Fullarton rehabilitated it in a weaker form in the mid-nineteenth century currency–banking debate, emphasizing the "law of reflux": notes lent out against real bills flow back to the issuing bank when the bills mature and are self-liquidating, so the supply of notes cannot permanently exceed the demand.

By the time the Federal Reserve Act was drafted, the real bills doctrine had achieved near-canonical status among American banking reformers. It reached the statute through an identifiable intellectual channel. Its leading academic champion was J. Laurence Laughlin of the University of Chicago, and it was Laughlin's student *Henry Parker Willis* who served as the principal technical expert to Representative Carter Glass and drafted much of the bill's language [Mehrling, 2014]. Laughlin and Willis held "that the monetary liabilities of the banking system should ideally be backed by holdings of short-term self-liquidating commercial bills," and they wrote that conviction into the eligible-paper regime of Section 13, confining the new system to the passive rediscount of commercial paper and away from "long-term investments and speculative capital markets" [Mehrling, 2014]. Their vision was contested: Paul Warburg, architect of the earlier Aldrich Plan, wanted an actively managed central bill market on the European model rather than the self-regulating accommodation the real-bills men favored, and Federal Reserve practice under Benjamin Strong would later drift toward Warburg's conception even though the Act's *text* followed Laughlin, Willis, and Glass [Mehrling, 2014].

Whether this made the Fed's doctrine genuinely Smithian or a homegrown American invention is a question on which the literature divides, and the division mirrors the antebellum ambiguity traced in Section 2. As Federal Reserve Bank of Richmond economist Robert Hetzel (2014) documented, "The founders of the Federal Reserve desired to end financial panics . . . [T]hey understood financial panics as resulting from speculative excess, especially on Wall Street. These 'real bills' views of financial panic originated in the 19th century American experience." On this reading the doctrine was less an inheritance from Smith than a distillation of decades of American panic experience; the Smithian lineage supplied its intellectual respectability. Either way, the Fed's founders reached for the *commercial-asset* half of Smith's system while paying far less heed to the *convertibility* half that had disciplined it—the same imbalance that had distinguished sound from unsound real-bills banking before the Civil War.

5.4 Congress's Dual Commitment in 1913

Parintha Sastry (2018) has argued that in its original passage the 1913 Act reflected a "steadfast commitment" to the real bills doctrine visible in two interrelated ways: first, by limiting the assets the Fed could purchase, discount, and accept as collateral to eligible commercial paper; second, by keeping government-sponsored credit enterprises separate from the Federal Reserve System. The Fed was to be a lender of last resort to *commercial* banking, not a financier of speculative or investment activity.

This limitation answered the rural critics of the Aldrich Plan who had feared that a central bank would become the instrument of Wall Street finance. By confining eligible paper to short-term, self-liquidating commercial loans, the drafters—Congressman Carter Glass and Senator Robert Latham Owen—sought to ensure that Fed credit would flow to Main Street merchants and farmers, not to stock-market speculators. Wilson secured the passage of the bill in the Senate by convincing Bryan's supporters that Federal Reserve notes, being obligations of the government, would satisfy the Populist demand for an "elastic currency" without replicating either the bond-backed rigidity of the National Banking System or the Wall Street favoritism of the Aldrich Plan.

5.5 The Echo of Earlier Reform Proposals

It is worth pausing to hear the echo of the pre-1908 reform proposals in the structure of the 1913 Act. The asset-currency advocates of the 1880s and 1890s had argued that notes should be backed by productive assets rather than government bonds; by 1913, the drafters went further—they required not merely that bank assets *back* the currency but that the specific assets eligible for Fed discount be active claims on ongoing commercial production. The warehouse-receipt proposals of agrarian reformers found a close analogue in the six-month agricultural paper carve-out in Section 13. The railroad-bond proposals did *not* survive: investment securities of any kind were explicitly excluded from eligible paper. The final legislation represented a deliberately narrow conception of what the currency should be "backed by"—narrower even than the Aldrich–Vreeland emergency currency, which had allowed almost any security as backing.

6 Post-1913 Controversies: Real Bills, Elasticity, and Instability

6.1 The Inflationary Bias

Critics of the real bills doctrine pointed to its inflationary potential almost from the opening of the Federal Reserve banks. James Parthemos of the Federal Reserve Bank of Richmond wrote in 1988:

> "This so-called commercial loan theory or real bills doctrine was a basic principle underlying the money functions of the new system. The essential fallacy in the doctrine was that note issue would also vary with the price level as well as the real volume of trade. Thus its operation would be inherently inflationary or deflationary." [Parthemos, 1988]

Friedman and Schwartz (1963) noted the same flaw, though without exploring its full implications.

The pro-cyclical logic was stark. In a boom, prices rise, the nominal value of commercial transactions increases, demand for discount rises, and the Fed—if mechanically following the real bills criterion—expands credit. In a contraction, the reverse occurs, amplifying the downturn.

6.2 The Great Depression and "Direct Pressure"

The most consequential invocation of the real bills doctrine came in late 1929. Long-serving Federal Reserve Board member Adolph C. Miller launched what Humphrey and Timberlake (2019) called the "Direct Pressure" initiative: a directive to all member banks requiring any bank seeking discount window assistance from the Fed to certify that it had never made speculative loans—especially loans for stock-market purposes. Under the real bills doctrine, speculative finance was categorically different from legitimate commercial lending, and the Fed was to be the commercial system's lender of last resort, not a backstop for speculators.

The result was catastrophic. Bankers, unwilling to submit to the implied interrogation (and doubting that the "hard-boiled" Fed would grant relief anyway), refrained from borrowing reserves even in the face of withering deposit outflows. Instead of discounting at the window, banks failed—nearly 9,000 of them. Their failure produced the one-third contraction of the money stock that Friedman and Schwartz held responsible for the Great Depression.

6.3 The Sargent–Wallace Reconsideration

In a 1982 article, Thomas Sargent and Neil Wallace (1982) reexamined the real bills doctrine in the context of overlapping generations models. They showed that within such models the real bills prescription—unfettered private intermediation—could be Pareto superior to the quantity-theoretic prescription, which imposes restrictions on private intermediation designed to separate "money" from credit. David Laidler (1984) countered that the Sargent–Wallace version of real bills differed substantially from the historical doctrine that had guided Fed policy: the historical real bills criterion was specifically about the *type* of collateral (short-term commercial paper versus speculative finance), not simply about whether intermediation should be restricted.

6.4 Legacy

The real bills provisions of the original Federal Reserve Act were gradually eroded over the following decades. The Banking Acts of 1932 and 1933, passed during the Depression, expanded the range of assets the Fed could discount; the 1935 amendments transferred power decisively to the Board of Governors and weakened the operational role of the reserve banks in policing eligible paper. Section 13(3), added to the Federal Reserve Act in 1932 and invoked extensively during the 2007–2009 financial crisis, authorizes the Fed to lend to any "individual, partnership, or corporation" in "unusual and exigent circumstances." The provision traces its legislative DNA to the Aldrich–Vreeland emergency currency mechanism of 1908, and stands at the furthest possible remove from the strict real bills criterion of the original Act.

7 Conclusion

The line from the National Banking Acts of 1863–1864 to the Federal Reserve Act of 1913 is coherent and traceable. The bond-backed currency of the national banking era was safe but rigid; the Comptroller of the Currency guaranteed the safety by inspecting bond portfolios, but no institutional mechanism existed to make the currency responsive to the needs of trade. Four decades of complaint—from Greenbackers, silver men, asset-currency advocates, and mainstream bankers alike—converged on the diagnosis of "inelasticity."

The Aldrich–Vreeland Act of 1908 was a halfway house: it broadened the permissible backing for emergency notes beyond government bonds, pointed toward the eventual solution, and created the intellectual infrastructure (the National Monetary Commission) that would produce the 1913 Act. The Federal Reserve Act then resolved the inelasticity problem by substituting the *real bills* criterion for the bond-backing requirement. Instead of asking "Is this note backed by a government bond?" the new regime asked "Is this note backed by a short-term, self-liquidating commercial claim?"

In choosing the real-bills criterion, however, the 1913 Act was not inventing a new principle but reviving an old one. As Section 2 argued, antebellum America had already contained two rival answers to the backing question: the bond-collateral principle of New York's free-banking law of 1838, which the National Banking Acts nationalized, and the commercial-asset (real-bills) principle, descended from Adam Smith and embodied most purely in the Louisiana Banking Act of 1842. The half-century of bond-backing from 1863 to 1913 was, in this longer view, an interruption; the Federal Reserve Act swung the country back toward the commercial-asset tradition that had governed the practice of the two Banks of the United States and the letter of Louisiana law. What it did *not* revive was the discipline that had made that tradition safe. Louisiana's real bills worked because a one-third specie reserve stood behind them; Smith's real bills were bounded because notes were convertible into gold on demand. The Fed inherited the commercial-asset half of the doctrine but, once the gold standard was suspended and then subordinated, operated it without the convertibility anchor— and reaped precisely the instability that Henry Thornton had predicted a century before.

The real bills criterion was thus simultaneously a monetary theory (the money supply ought to expand and contract with productive activity), a political compromise (Fed credit should serve Main Street, not Wall Street), and an echo of a debate over what money should be "backed by" that stretched back through Louisiana 1842 to *The Wealth of Nations*. Its subsequent history—above all its disastrous role in the onset of the Great Depression—showed that the doctrine embedded in 1913 carried its own serious defects. But understanding how it came to be embedded requires tracing the full arc of American monetary thought from Adam Smith and the antebellum banks through the Civil War to the Progressive Era, a story whose protagonists include not only the drafters of legislation but the farmers, bankers, and economists whose decades of argument about what should back the currency shaped the institutional response.

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